

NEWPORT NEWS DOMINION TERM, VA, USA

WMO: 997782

Lat: 36.962N

Lon: 76.424W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.4	-3.4	N/A	N/A	N/A	N/A	N/A	N/A	13.5	1.7	12.1	5.3	4.7	0	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	32.0	N/A	30.8	N/A	29.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	10.0	8.6	-7.4	34.6	2.6	1.7	-9.3	35.8	-10.8	36.8	-12.3	37.7	-14.2	38.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	4.6	5.8	9.2	15.5	20.3	24.9	26.8	25.8	23.2	18.0	11.2	8.4
	DBStd	8.58	4.96	4.55	4.55	3.94	3.39	2.63	1.96	1.77	2.58	3.67	3.73	4.43
	HDD10.0	498	177	130	72	4	0	0	0	0	0	1	30	83
	HDD18.3	1754	425	349	284	102	21	1	0	0	1	49	214	309
	CDD10.0	2759	11	14	47	168	318	446	519	491	397	249	66	33
	CDD18.3	974	0	0	1	16	80	196	261	232	148	39	1	1
	CDH23.3	6952	0	0	1	39	313	1471	2446	1832	759	90	0	0
	CDH26.7	1664	0	0	0	2	48	386	734	379	109	6	0	0
Wind	WSAvg	4.1	4.3	4.4	4.2	4.4	4.2	4.1	4.2	3.6	3.7	4.0	3.8	4.1
Precipitation	PrecAvg	1207	82	76	93	83	90	107	139	135	137	100	80	84
	PrecMax	1534	149	218	251	153	135	263	240	345	444	207	280	172
	PrecMin	782	28	24	16	15	34	42	33	40	17	7	4	34
	PrecStd	205	32	41	44	34	26	50	50	74	98	53	59	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.3	19.2	21.9	26.2	30.4	33.3	34.3	32.2	30.5	27.4	20.6	20.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	16.0	16.7	19.2	24.4	28.0	31.4	32.4	30.8	29.1	25.4	19.0	18.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	13.9	14.8	17.7	22.8	26.5	30.1	31.2	29.9	28.0	24.2	17.8	16.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	12.0	12.7	16.0	21.2	25.2	29.0	30.1	28.7	26.8	23.0	16.7	14.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.5	6.5	6.7	6.2	5.8	5.6	4.8	4.7	5.3	5.7	5.8
		MCDBR	8.8	9.1	9.0	8.6	7.7	7.4	7.0	6.5	6.0	6.0	6.5	7.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.307	0.320	0.347	0.392	0.435	0.486	0.513	0.507	0.408	0.356	0.324	0.312	
	taud	2.538	2.498	2.453	2.340	2.249	2.159	2.097	2.100	2.369	2.509	2.547	2.558	
	Ebn at Noon	889	918	919	890	852	806	780	775	844	865	862	860	
	Edh at Noon	78	91	104	124	138	151	159	156	113	89	77	72	
All-Sky Solar Radiation	RadAvg	2.35	3.11	4.13	5.26	5.92	6.36	6.05	5.46	4.44	3.53	2.66	2.10	
	RadStd	0.19	0.34	0.33	0.39	0.52	0.41	0.30	0.30	0.37	0.41	0.22	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+107

Nomenclature: See separate page